Efficient Solar Filament Segmentation at Ensemble Parity

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

Solar filaments are thin, low-contrast chromospheric structures whose instance segmentation is dominated by annotator disagreement and by fragmentation or over-merging errors. We describe a compact ConvNeXt-atto U-Net (4.28 M parameters) trained with knowledge distillation, randomized-annotator targets and stochastic weight averaging, then deployed with single-pass 2048-pixel inference, eight-view dihedral test-time augmentation and a decoupled "KEEP" instance-acceptance rule. On the 107-file validation split the student reaches an exact multi-annotator Panoptic Quality (PQ) of 0.4591 and a public leaderboard score of 0.38—equal to a 171 M four-model ensemble—while using 107 GMAC and 103 ms per image (40× fewer parameters and ∼30× fewer MACs at equal leaderboard score). Under the official IoU> 0.5 matching, the submitted system produces no one-to-many or many-to-one relations; at IoU> 0 only 2.4% of ground-truth instances and 5.1% of predictions have multiple partners. The decoupled KEEP rule alone is worth +0.0158 nested-CV PQ and moved the leaderboard from 0.35 to 0.37 with no retraining; full-disk inference and TTA added the final quantum to 0.38.

CCS Concepts

• **Computing methodologies** → **Image segmentation**; *Supervised learning*.

Keywords

solar filaments, instance segmentation, panoptic quality, knowledge distillation, model efficiency

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

Our contribution is threefold. (i) We show that the binding constraint of the previous system was not model capacity but a coupled post-processing rule: decoupling the contour threshold from instance acceptance yields +0.0158 nested-CV PQ and +0.02 leaderboard without retraining. (ii) A 4.28 M-parameter distilled student, fine-tuned with randomized-annotator targets and SWA, matches a 171 M four-model ensemble at PQ 0.38. (iii) The resulting pipeline is single-pass, deterministic and reproducible; its weights, inference notebook and verification scripts are released.

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026 2026.

2 Methodology

Data and split. MAGFiLO contains 707 GONG H-α full-disk images (2048 × 2048) with 1–3 expert annotators per image (1,154 records; 8,199 instance masks). We split *files*, not annotator records, 85/15 (600 train / 107 validation), so no image appears in both sets through a different annotator. The exact metric is computed against every annotator's mask with all-pairs IoU> 0.5 matching.

Preprocessing. Images are read as raw 8-bit grayscale. At inference the solar disk is detected on a 4× downsample (threshold, morphological close, convex hull, erosion) and used only as an outside-disk gate. No external data is used at inference: the gate is derived from the test image itself. The single-channel image is replicated to three channels and normalized on the GPU with ImageNet mean/std, matching the pretrained stem. No CLAHE or radial flattening is used; both were falsified on this data.

Model and training. The student is a ConvNeXt-atto encoder [5] with a four-stage U-Net decoder [6] (channels 128/64/32/16), 4.28 M parameters. It is trained for 30 epochs on 1024-pixel crops (batch size 4, ≈5 min per epoch on one GPU) with a balanced loss 0.5 BCE + 0.5 Tversky($\alpha = \beta = 0.5$), AdamW (lr 10^{-4}), linear warmup followed by cosine annealing, and per-batch EMA (raw weights are used for inference; EMA loses instance PQ). Targets are the first annotator's mask; a six-epoch recall fine-tune switches to *random_annotator* (all 1,154 records), starts from the complete-GT checkpoint, uses knowledge distillation from a five-teacher ensemble (MSE, weight 5) [3], and averages the last three epochs (SWA). The fine-tune selects checkpoints by full-image PQ under the deployed rule, not by pixel Dice.

Inference and post-processing. Fig. 1 shows the end-to-end path. The network is fully convolutional, so the whole 2048 × 2048 image is processed in one forward pass (107 GMAC, 103 ms FP16), optionally with eight dihedral views (D4 TTA8) whose sigmoid maps are averaged. We then apply the decoupled KEEP rule: form 8-connected components of $p > t_{\text{lo}}$, and accept a component iff

$$\text{peak}(p) > t_{\text{hi}}, \qquad \text{mean}(p) \geq m_{\text{p}}, \qquad \text{area} \geq a_{\text{min}}.$$

The submitted student (S3) uses $t_{\text{lo}} = 0.35$, $t_{\text{hi}} = 0.99$, $m_{\text{p}} = 0.85$, $a_{\text{min}} = 100$. Components are exported as COCO RLE. The nested-CV optimum was $a_{\text{min}} = 300$; the two are indistinguishable on validation (0.45909 vs 0.45906), and the submitted CSV uses 100. The accuracy reference (S2) uses a 171 M four-model 1024-pixel ensemble with $t_{\text{lo}} = 0.35$, $t_{\text{hi}} = 0$, $m_{\text{p}} = 0.80$, $a_{\text{min}} = 100$; the leanest point (S4) uses a 64-wide decoder and $t_{\text{lo}} = 0.30$, $t_{\text{hi}} = 0.99$, $m_{\text{p}} = 0.85$, $a_{\text{min}} = 100$.

3 Evaluation

Protocol. We use the exact competition metric: multi-annotator per-image ground truth and all-pairs IoU> 0.5 matching, giving

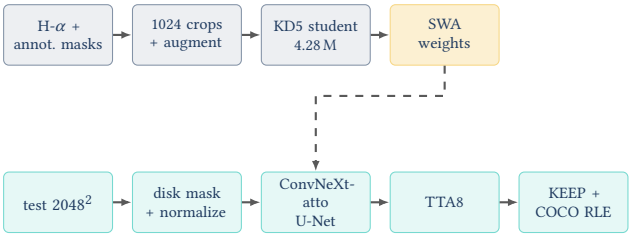


Figure 1: End-to-end pipeline. Navy: training (once), with KD5 teachers and SWA. Teal: inference per test image; the gold SWA weights initialize the U-Net. The released notebook regenerates the submission from the weights without retraining.

System	Params	MACs	Val PQ	LB
S2 ensemble (4×1024, TTA8)	171.0 M	3.23 TMAC	**0.4773**	**0.38**
S3 student (submitted)	**4.28 M**	**107 GMAC**	0.4591	**0.38**
S4 student (64-wide, leanest)	3.75 M	64.8 GMAC	0.4599	0.37

Table 1: Validation PQ (107 files, exact metric) and public leaderboard. MACs are one forward per model, not multiplied by the eight TTA views.

$\mathrm{PQ} = \frac{\sum_{(p,g)\in TP} \mathrm{IoU}(p,g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}$ [4]. Rule and ensemble choices are selected with nested 5-fold cross-validation on the 107 validation files and compared with a paired file-level bootstrap (4,000 resamples).

Results. Table 1 summarises the three operating points. The submitted student reaches 0.4591 validation PQ and 0.38 on the public leaderboard, equal to the 171 M ensemble, at 40× fewer parameters and about 30× fewer MACs on the same single-pass path. The 3.75 M leanest point is 0.0008 better in validation but lands one leaderboard quantum lower (0.37), which is within the public-subset noise floor. The historical 0.35 plateau was therefore a decision-rule and inference-path problem, not a model-capacity ceiling.

Cost. Fig. 3 places the three operating points on the cost–quality plane. MACs are analytic for one 2048^2 forward and therefore do not depend on the measurement hardware; marker area encodes parameter count. The submitted student sits 30× left of the ensemble at the same public score, and the leaner S4 buys a further 39% of MACs for one leaderboard quantum. Wall-clock on one GPU follows the same ordering: 103 ms per image single-pass for S3, against 278 ms for the 8×1024 sliding window at equal PQ.

Accuracy distributions. On the 933 matched validation pairs, mean IoU is 0.689 (median 0.695) and mean Dice is 0.813 (median 0.820); Fig. 4 shows the distributions. The matched-pair mass is concentrated at high overlap, consistent with a contour that preserves thin barbs rather than smoothing them.

Fragmentation and over-merging. Under the official IoU> 0.5 matching, the submitted system has *zero* one-to-many and many-to-one relations: every matched ground-truth instance has exactly one prediction and vice versa. At the looser IoU> 0 diagnostic (Fig. 4, lower left), 2.4% of ground-truth instances overlap more than one prediction and 5.1% of predictions overlap more than one ground-truth instance. The KEEP rule is what removes the residual fragmented/merged candidates before they reach the metric.

Effect (nested OOF)	ΔPQ	95% CI
KEEP vs coupled	+0.0158	[0.0087, 0.0231]
tiny + randann	+0.0127	[0.0045, 0.0217]
tiny + randann + SWA	+0.0105	[0.0032, 0.0177]
dec128 + randann + SWA	+0.0051	[−0.0056, 0.0167]

Table 2: Paired file-level bootstrap (4,000 resamples) for the rule and fine-tune deltas. All intervals are nested 5-fold OOF.

Error anatomy. At the deployed rule S3 has 933 matched instances, 474 false positives and 461 false negatives. The FNs decompose into 292 missed (63.3%), 163 boundary (35.4%) and 6 merged (1.3%). Moving from the old coupled rule to KEEP on the same TTA8 probabilities changes TP/FP/FN from 916/575/478 to 933/474/461, raising RQ from 0.635 to 0.666 while SQ moves from 0.686 to 0.689: the rule adds 17 true positives and removes 101 false positives. Post-acceptance dilation and erosion are both negative (dil1 −0.025, dil2 −0.094, ero1 −0.022 PQ), so the residual is not a thin-boundary problem.

Member quality is not the ensemble's constraint. The recall fine-tune is worth +0.0127 nested-OOF PQ on its own (95% CI [0.0045, 0.0217]), yet adding it to the four-model ensemble at weights of 0.10, 0.15 and 0.20 moves PQ by +0.0003, −0.0001 and +0.0002, and adding the distilled student at weight 0.10 gives +0.0001 (−0.0024 to +0.0026); every interval spans zero. A member can be materially better alone and worth nothing to the ensemble when the residual error is shared: all members miss the same unlabelled structures. Ensemble design is therefore also saturated.

Strengths and weaknesses. The system is fast, small, and free of explicit fragmentation/merging under the official matching. Its weakness is recall: 63.3% of remaining FNs are pure misses and 35.4% are boundary/agreement failures, and the labels themselves are noisy (inter-annotator union-mask IoU 0.467), so validation absolute levels are optimistic relative to the sparser test labels. The public leaderboard is quantised at 0.01 and cannot arbitrate sub-0.01 differences; we therefore report deltas and bootstrap intervals, not just absolute PQ. More positive training data (5-fold over all 707 images) is the most promising remaining lever.

Qualitative comparison. Fig. 2 shows zoomed validation examples. The prediction follows the elongated body and barbs in (a), the curved filament in (b), and one annotator's drawing in (e) and (f); in (c) and (d) the filament is missed outright rather than spuriously detected or merged. Limb artifacts are suppressed by the solar-disk gate.

Uncertainty and efficiency. Table 2 makes the uncertainty explicit: the KEEP rule and the tiny recall fine-tune are significant; the dec128 fine-tune is not. Table 1 gives the efficiency: S3 is about 30× left of S2 at the same 0.38 LB, and S4 is the leanest point but one LB quantum lower.

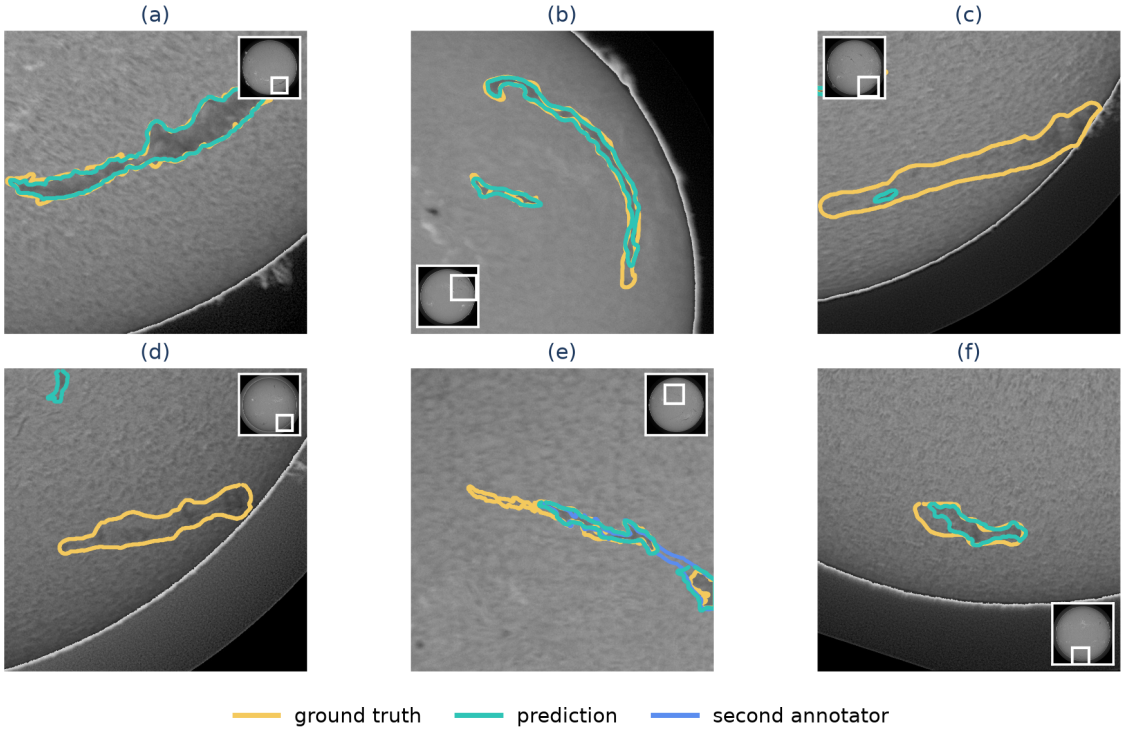


Figure 2: Zoomed validation examples. Gold: ground truth; teal: prediction; blue: a second annotator. (a,b) the prediction follows thin barbs and a large curved filament; (c,d) two filaments are missed outright; (e,f) two cases where the annotators disagree and the prediction follows one of them. Insets show the full-disk location.

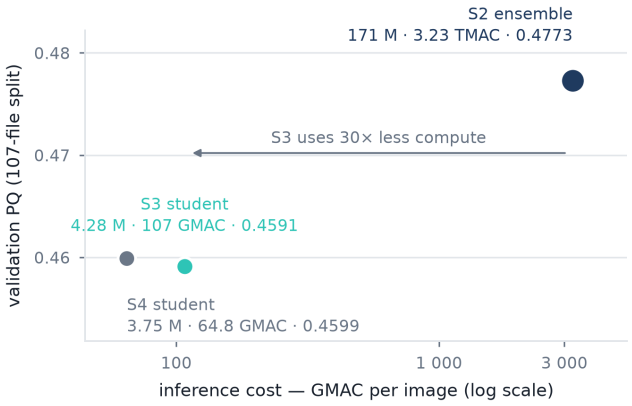
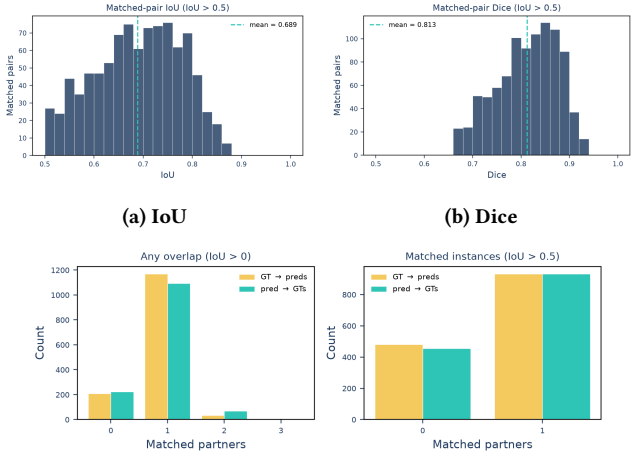


Figure 3: Inference cost against validation PQ. MACs are analytic for one 2048^2 forward, so the comparison does not depend on the measurement hardware; marker area encodes parameter count. S3 sits 30× left of S2 at the same public score, and S4 a further 39% left for one leaderboard quantum.



(a) IoU **(b) Dice**

(c) Fragmentation / over-merging relations

Figure 4: Matched-pair IoU/Dice and one-to-many/many-to-one relations for S3 on the 107 validation files. The lower subfigure counts matched partners at IoU> 0 (left) and under the official IoU> 0.5 matching (right).

Reproducibility. The submitted package contains the final weight (16.4 MB), a requirements file, a 13-check submission verifier, and a notebook that runs the whole pipeline: file split, preprocessing, model, optional training, inference, KEEP post-processing, submission and validation scoring. On this hardware the notebook

reproduces the reported validation PQ (0.4591) exactly on the same 107 files, and the submitted CSV (1,279 rows over 177/180 test images, the three empty images legitimate, all checks passing) on 1,277 of its 1,279 rows byte-for-byte. That difference is one image, where two contours move by a pixel and a third boundary component passes the area test, giving 1,280 rows. The stored FP16 test cache is not a substitute for re-inference: it differs from the submitted file on 31 rows across 22 images. Re-inference takes about a minute on one GPU.

Limitations. Two caveats bound these numbers. The validation labels are materially noisier than the test labels: 27% of confident detections match no annotator at all, and 25–35% of the remaining false negatives are boundary disagreements rather than misses, so absolute validation PQ is optimistic and only deltas are portable. The public leaderboard, in turn, is quantised at 0.01 and cannot resolve the effects we report; the accuracy ensemble is therefore selected on its validation PQ (0.4773), not on the leaderboard. The largest untested lever is data coverage: 5-fold training over all 707 images, rather than architectural change.

Code quality. Evaluation and submission share a single model registry, so a tuned rule cannot drift from the deployed one; probability maps are cached per inference path, which makes every rule and ensemble comparison in this report re-runnable on the same 107 files without a GPU; and the exact multi-annotator metric has one implementation, used by both the tuning sweeps and the figures. A 13-check verifier validates each submission CSV against the official format before upload.

4 Acknowledgment

The Filament Segmentation Challenge 2026 is sponsored by the U.S. National Science Foundation (NSF) National Solar Observatory (NSO) and its organization is partially supported by NSF grants 2209912, 2433781 and 2511630 (CISE/CSE, OAC, and AST/MPS).

This work utilizes GONG data obtained by the NSO Integrated Synoptic Program, managed by the National Solar Observatory, which is operated by the Association of Universities for Research in Astronomy (AURA), Inc. under a cooperative agreement with the National Science Foundation and with contribution from the National Oceanic and Atmospheric Administration. The GONG network of instruments is hosted by the Big Bear Solar Observatory, High Altitude Observatory, Learmonth Solar Observatory, Udaipur Solar Observatory, Instituto de Astrofísica de Canarias, and Cerro Tololo Interamerican Observatory.

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